APERIO | AUTONOMOUS EARTH INTELLIGENCE PLATFORM

AUTHORITY DISASTER INTELLIGENCE ASSESSMENT • FLOOD

Location: Tinsukia, Assam, India • Generated: 09 Oct 2026 05:51 UTC • Status: AWAITING HUMAN REVIEW

Target Event	Flood • Tinsukia	**Risk Probability**	**26% (HIGH)**
Affected Area	N/A	**Model Confidence**	88% Indicative
Exposed Population	Assessed in loop	**Antecedent Rainfall**	120 mm (72h: 572mm)
Terrain Slope	15° (Steep escarpment)	**SAM Scar Boundary**	Validated (7.03 km²)
Satellite Observation	Sentinel-1 SAR + Sentinel-2 MSI	**Human Verification**	AWAITING HUMAN REVIEW

Multi-Agent Intelligence & Evidence Progression

1. Satellite Ingestion: Copernicus Sentinel-1 C-SAR IW mode acquired observation passes over Tinsukia. Radar backscatter penetration enabled continuous surface change mapping.
2. DEM Terrain Modeling: Horn algorithm computed local terrain slope of 15° across drainage lines.
3. SAM Spatial Segmentation: Delineated spatial impact boundary encompassing high-risk terrain.
4. XGBoost & Bayesian Sequential Update: Domain evidence likelihoods updated sequential prior belief to yield calibrated posterior risk of **26%**.

Response Recommendations & Relocation Planning

Designation	Capacity / Occupancy	Distance / Road Status	Hazard Exposure
Designated Relief Center (Tinsukia)	650 / 380 (270 Available)	2.4 km • Open	**LOW RISK**
Community Relief Hall West	420 / 310 (110 Available)	4.1 km • Standby	**MODERATE**
District Camp North	800 / 220 (580 Available)	6.8 km • Clear	**LOW RISK**

Satellite Change Intelligence & Extent Quantification

Analyzed Extent	12056.04 km² (Full Granule)	**Methodology**	Visual Water Index (VWI) & Optical Surface Absorption Delta
Baseline Affected Extent	4470.56 km² (2024-05-22)	**Sensor & Product**	Sentinel-2 MSI Level-2A (Tile T46RGR)
Latest Affected Extent	3731.22 km² (2024-07-31)	**Cloud Cover**	Pre: 0.04% • Post: 13.15%
Newly Affected Area	**523.32 km²**	**Net Change**	**-739.34 km²** (-16.5%)
Detection Zones	6 Delineated Zones	**Integrity Status**	**CANDIDATE_CHANGE**

Accountable Operator Review & Audit Trail

Reviewing Desk: APERIO Response Operations Desk
Operator Notes: Automated verification report

Operational Boundary: Decision Support Output only. All evacuations require local ground confirmation.